

Tong Xu

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EDUCATION

George Mason University

Ph.D. in Computer Science

Fairfax, USA

Aug. 2023 - Present

University of Southern California

M.S. in Computer Science

Los Angeles, USA

Aug. 2021 - May 2023

Nanjing University of Information Science & Technology

B.E. in Network Engineering

Nanjing, China

Sep. 2017 - Jun. 2021

RESEARCH STATEMENT

Research Interests: My research focuses on *motion planning, robot learning, world models, and embodied intelligence*, with an emphasis on learning from large-scale interaction data for real-world robotic systems.

Research Questions: How can robots learn *structured, transferable representations of physical interaction* for *generalizable prediction, planning, adaptation, and control* across diverse embodiments and environments?

RESEARCH EXPERIENCE

George Mason University

Research Assistant, [RobotiXX Lab](#)

Advisor: [Xuesu Xiao](#)

Fairfax, USA

Aug. 2025 - Present,

Summer 2024 - 2026

Verti-Bench: A General and Scalable Off-Road Mobility Benchmark

- Created a general off-road mobility benchmark *Verti-Bench* on vertically challenging terrain with 100 off-road environments and 1000 navigation tasks scalable to various vehicle types
- Incorporated millions of off-road terrain features including geometry, semantics (rigid and deformable), and obstacles
- Provided various datasets and a RL interface to facilitate data-driven off-road mobility
- Implemented ten different off-road mobility systems evaluated in the benchmark

Verti-WM: A Physics-Aided Exteroceptive World Model for Off-Road Reinforcement Learning

- Proposed a physics-aided exteroceptive world model that recurrently fuses Transformer-based and neuro-symbolic kinodynamic specialists for terrain-conditioned 6-DoF rollouts across rigid and deformable terrain
- Enabled reinforcement learning entirely within the learned world model through map-conditioned autoregressive rollouts, eliminating further access to the high-fidelity simulator during policy optimization
- Reduced prediction error by up to 34.6% and 21.7% over data-driven and physics-based baselines, while accelerating policy training by 23.6X and achieving an 80% real-world success rate

VertiAKD: Adaptive Off-Road Kinodynamics on Vertically Challenging Terrain

- Proposed a shared mobility representation that jointly encodes vehicle configurations, trajectory transitions, and local terrain features to capture kinodynamic (dis)similarity across vehicle-terrain interactions
- Designed a terrain-aware function encoder that models forward kinodynamics with neural Ordinary Differential Equation (ODE) basis functions incorporating elevation and semantic information, initialized by a coefficient prior constructed from relevant mobility descriptors in the shared latent space
- Developed an online adaptation scheme that refines the transferred coefficient prior from streaming observations using recursive least squares, without gradient-based retraining during deployment
- Reduced long-horizon prediction error by up to 34.52% over direct mobility descriptor transfer across diverse unseen vehicle configurations and 94.43% over competing baselines

VertiAdaptor: Online Kinodynamics Adaptation for Vertically Challenging Terrain

- Presented a unified terrain representation using the Sliced Wasserstein Autoencoder (SWAE) to efficiently obtain elevation and semantic embeddings beneath the vehicle

- Proposed the first $\mathbb{SE}(3)$ kinodynamics online adaptation framework based on complex off-road elevation and semantics that enables rapid kinodynamic model updates through least-squares optimization
- Improved prediction accuracy by up to 23.9% and achieved a 5X faster adaptation time compared with Multi-Layer Perceptron fine-tuning, Model-Agnostic Meta-Learning, and neural ODE

VertiSelector: Automatic Curriculum Learning (ACL) for Wheeled Mobility

- Presented the *VW-Chrono* simulator, designed for wheeled mobility on vertically challenging terrain to algorithmically generate varied vertically challenging terrain for ACL
- Proposed a novel ACL framework that samples training terrain based on estimates of future learning potential (TD-error)
- Combined Proximal Policy Optimization (PPO) with the SWAE structure to efficiently learn wheeled mobility in *VW-Chrono*
- Improved navigation performance by 23.08% in terms of success rate compared against a manually designed curriculum, vanilla RL, a hybrid (classical and learning) method and two classical baseline approaches

Boston University

Research Intern, [H2X Lab](#), Advised by *Eshed Ohn-Bar*

- Research Topics: deep visual odometry, motion estimation

Boston, USA

May 2022 - Aug. 2022

Nanjing University of Information Science & Technology

Research Assistant, Advised by *Xiaorui Zhang*

- Research Topics: domain adaptation, emotion recognition

Nanjing, China

Oct. 2019 - May 2021

PUBLICATIONS

(* indicates equal contribution)

Journal

- [J4] X. Xiao, Z. Xu, S. A. Ghani, A. Datar, D. Song, P. Stone, A. Mazen, K. Yazdipaz, I. Mateyaunga, M. Faied, Y. Lu, **T. Xu**, N. Mahammad, W. Kim, J. Reasoner, and N. Bezzo. Autonomous Ground Navigation in Highly Constrained Spaces: Lessons Learned from The Forth BARN Challenge at ICRA 2025. *IEEE Robotics & Automation Magazine (RA-M)*, Vol. 33, Iss. 1: 186-196, March 2026.
- [J3] Z. Hu, **T. Xu**, X. Xiao, and X. Wang. CARoL: Context-aware Adaptation for Robot Learning. *IEEE Robotics and Automation Letters (RA-L)*, Vol. 10, No. 11: 12063-12070, November 2025.
- [J2] X. Cai, J. Queeney, **T. Xu**, A. Datar, C. Pan, M. Miller, A. Flather, P. R. Osteen, N. Roy, X. Xiao, and Jonathan P. How. PIETRA: Physics-Informed Evidential Learning for Traversing Out-of-Distribution Terrain. *IEEE Robotics and Automation Letters (RA-L)*, Vol. 10, No. 3: 2359-2366, March 2025. ([Video](#))
- [J1] X. Zhang, **T. Xu**, W. Sun, and A. Song. Multiple source domain adaptation in micro-expression recognition. *Journal of Ambient Intelligence and Humanized Computing*, Vol. 12, No. 8: 8371–8386, 2021.

Conference

- [C10] **T. Xu**, C. Pan, A. Datar, and X. Xiao. VertiAdaptor: Online Kinodynamics Adaptation for Vertically Challenging Terrain. *Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [C9] C. Pan, A. Datar, A. Pokhrel, M. Choulas, M. Nazeri, **T. Xu**, and X. Xiao. Traverse the Non-Traversable: Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain. *Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [C8] Y. Lu, M. Mao, **T. Xu**, L. Wang, X. Lin, and X. Xiao. Adaptive Dynamics Planning for Robot Navigation. *Accepted by IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [C7] H. Chen, A. Datar, **T. Xu**, F. Cancelliere, H. Rangwala, M. B. Rao, D. Song, D. Eichinger, and X. Xiao. Verti-Arena: A Controllable and Standardized Indoor Testbed for Multi-Terrain Off-Road Autonomy. *IEEE International Symposium on Safety Security Rescue Robotics (SSRR)*, pp. 133-138, October 2025. ([Video](#))
- [C6] L. Wang*, **T. Xu***, Y. Lu, and X. Xiao. Reward Training Wheels: Adaptive Auxiliary Rewards for Robotics Reinforcement Learning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 15262-15267, October 2025. ([Video](#))

- [C5] **T. Xu**, C. Pan, and X. Xiao. VertiSelector: Automatic Curriculum Learning for Wheeled Mobility on Vertically Challenging Terrain. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 11136-11143, October 2025. ([Video](#))([Code](#))
- [C4] Y. Lu, **T. Xu**, L. Wang, N. Hawes, and X. Xiao. Incremental Dynamics Planning for Robot Navigation. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4559-4565, October 2025.
- [C3] **T. Xu**, C. Pan, M. B. Rao, A. Datar, A. Pokhrel, Y. Lu, and X. Xiao. Verti-Bench: A General and Scalable Off-Road Mobility Benchmark for Vertically Challenging Terrain. *Robotics: Science and Systems (RSS)*, June 2025. ([Website](#))([Video](#))([Code](#))
- [C2] **T. Xu**, C. Pan, and X. Xiao. Reinforcement learning for wheeled mobility on vertically challenging terrain. *IEEE International Symposium on Safety Security Rescue Robotics (SSRR)*, pp. 125-130, November 2024.
- [C1] **T. Xu**. Single-view and multi-view methods in marker-less 3d human motion capture. *Journal of physics: conference series*, pp. 012022, 2019.

Preprint

- [P6] C. Pan*, **T. Xu***, and X. Xiao. Verti-WM: A Physics-Aided Exteroceptive World Model for Off-Road Reinforcement Learning. *under review*, 2026.
- [P5] B. Wang, **T. Xu**, D. Kosukhin, Y. Yeung, Y. Lu, and X. Xiao. Learning Scene-Aware Humanoid Locomotion through 3D Clutter from Immersive Human Demonstrations. *under review*, 2026.
- [P4] M. Ogunsina, A. Paudel, **T. Xu**, X. Xiao, and G. Stein. Determinization-Guided Planning for Multi-Robot Multi-Goal Navigation in Risky Environments. *under review*, 2026.
- [P3] C. Pan, **T. Xu**, F. Cancelliere, and X. Xiao. NeSAM: Neuro-Symbolic Kinodynamics with Soil Adaptation for Off-Road Mobility. *under review*, 2026.
- [P2] **T. Xu**, C. Pan, F. Cancelliere, and X. Xiao. VertiAKD: Adaptive Off-Road Kinodynamics on Vertically Challenging Terrain. *under review*, 2026.
- [P1] **T. Xu***, C. Pan*, and X. Xiao. CAR: Cross-Vehicle Kinodynamics Adaptation via Mobility Representation. *under review*, 2026.

HONORS

- Doctoral Research Scholarship - Tier 1, George Mason University (05/2026)
- IROS-SDC 2025 Travel Award (08/2025), IROS 2026 RAS Travel Grant (07/2026)
- 2nd Place, ICRA 2025 The Benchmark Autonomous Robot Navigation (BARN) Challenge (05/2025)
- The Outstanding Graduate of Jiangsu (06/2021)
- President's Scholarship of Nanjing University of Information Science & Technology (09/2020)
- China National Scholarship for Undergraduates (top 1%) (09/2020)

ACADEMIC SERVICE

Journal Reviewer: T-RO, RA-L, RA-M, RA-P, TVT

Conference Reviewer: RSS, ICRA, IROS, CDC, SSRR, UR-RAD, CCTA

Graduate Teaching Assistant

CS 485: [Autonomous Robotics](#)

CS 262: Introduction to Low-Level Programming

CS 112: Introduction to Computer Programming

George Mason University

Spring 2025

Spring, Fall 2024

Fall 2023

SKILLS

Programming: Python, C++, LaTeX, Java, Node.js, SQL, Linux

Frameworks: Pytorch, Stable-Baselines3, NumPy, Anaconda, Git, Slurm, ROS1, ROS2

Robots: [Verti-Wheelers](#), Unitree G1, Unitree Go1, Unitree Go2